

Algorithm:

1. Push the root node in the Queue.
2. Loop until the queue is empty.
3. Remove the node from the Queue.
4. If the removed node has unvisited child nodes, mark them as visited and insert the unvisited children in the queue.

PROGRAM :

```
#include <stdio.h>
#include <conio.h>
int a[20][20], v[20], visited[20], n, i, j, f=0, r=-1;
void bfs (int v) {
    for (i=1; i<=n; i++)
        if (a[v][i] && !visited[i])
            a[++r] = i;
    if (f<=r) { visited[a[f]] = 1;
                bfs (a[f++]); }
}
void main () { int v; clrscr();
    printf("\n Enter the number of vertices: ");
    scanf ("%d", &n);
```



```
for (i=1; i<=n; i++) { a[i] = 0;
visited[i] = 0; }
printf("\n Enter graph data in matrix form: \n");
for (i=1; i<=n; i++)
for (j=1; j<=n; j++)
scanf("%d", &a[i][j]);
printf("\n Enter the starting vertex:");
scanf("%d", &v);
bfs(v);
printf("\n The node which are reachable are: \n");
for (i=1; i<=n; i++)
if (visited[i])
printf("%d\t", i);
else printf("\n bfs is not possible");
getch();
}
```

Output: